M4 (C+T+D), 12m window, LightGBM--Optuna | Relative Disturbance

Sites sorted by site-maximum relative disturbance; dashed lines = natural breaks, mean ± 0.5 SD (top to bottom: 21 High, 42 Mid, 39 Low)

GPP (g C m⁻² season⁻¹)

NEP (g C m⁻² season⁻¹)

ET (mm season⁻¹)

WUE (g C mm⁻¹)

US-NC2 ENF
BE-Maa CSH
DE-Akm WET
IT-BFt DBF
CA-DB2 WET
DE-Har ENF
CZ-RAJ ENF
IT-Cp2 MF
SE-Nor ENF
DE-Tha ENF
IT-Ren ENF
US-xRM ENF
US-MtB ENF
IT-Lav ENF
DE-RuC OSH
US-Vcp ENF
US-xJE ENF
CH-Dav ENF
CA-LU2 OSH
AU-APL WET
DE-Msr ENF
CN-Sb2 ENF
RU-Fyo ENF
ES-Hen ENF
US-Rpf DBF
AT-Zoe MF
AT-Mmg ENF
IL-Yat ENF
FI-Hyy ENF
CA-TP3 ENF
FI-Rk2 ENF
BE-Vie MF
IT-BsB DBF
FR-Bil ENF
CA-TP1 ENF
US-xDJ ENF
US-BZB WET
FI-Let ENF
US-CMW DBF
DE-Obe ENF
BE-Bra ENF
KR-Kw1 ENF
CZ-BK1 ENF
US-xUK DBF
JP-Yms DBF
FR-Pue EBF
US-xWR ENF
FR-FBn ENF
ES-Crg CSH
US-BZo WET
US-HB2 ENF
CH-Lae MF
EE-Stg ENF
CN-BeO MF
FI-Var ENF
FI-Ken ENF
US-xBL DBF
US-xBN ENF
US-HB3 ENF
US-MBP WET
DE-Hzd DBF
KR-Kw2 DBF
IT-SR2 ENF
SE-Svb ENF
CL-SDP WET
US-xSC DBF
RU-Fy2 ENF
US-SSH DBF
US-xBR DBF
DK-Sor DBF
CA-TPD DBF
CZ-Lnz DBF
DE-HoH DBF
CN-Dzo EBF
DE-Hai DBF
CZ-Stn DBF
JP-Api DBF
US-HBK DBF
US-xGR DBF
US-xDL MF
US-xML DBF
US-CRK ENF
NL-Loo ENF
SE-Ros ENF
US-xAB ENF
US-MOz DBF
US-CS2 ENF
US-RRC WET
SE-Htm ENF
IT-Noe CSH
US-Bar DBF
US-xUN MF
CL-ACF ENF
CN-Sb1 DNF
US-xTR DBF
US-UMd DBF
DK-Gds ENF
US-xST DBF
FR-Fon DBF
US-xSE DBF
US-xRN DBF
JP-Ynf EBF

IGBP

Relative Disturbance (%)
breaks 12.0 | 22.8
60
40
20
0

-300 -200 -100 0 100 200
Net signed disturbance SHAP (response units)

-200 -100 0 100
Net signed disturbance SHAP (response units)

-40 -20 0 20 40
Net signed disturbance SHAP (response units)

-1.5 -1.0 -0.5 0.0 0.5
Net signed disturbance SHAP (response units)